

CSA0503-DATABASE MANAGEMENT SYSTEMS

EXPERIMENT 6 - SELECT with various clause – BETWEEN, IN, Aggregate function

1) Find the average mark of “C002”

```
mysql> CREATE TABLE STUDENT7 (  
->     Reg_No INT PRIMARY KEY,  
->     Name VARCHAR(15),  
->     CourseNo VARCHAR(4),  
->     Mark INT  
-> );  
Query OK, 0 rows affected (0.07 sec)  
  
mysql> INSERT INTO STUDENT7 VALUES  
-> (101, 'Arun', 'C002', 80),  
-> (102, 'Divya', 'C002', 70),  
-> (103, 'Karthik', 'C001', 90),  
-> (104, 'Mohan', 'C002', 60),  
-> (105, 'Priya', 'C003', 75),  
-> (106, 'Ram', 'C002', 90);  
Query OK, 6 rows affected (0.02 sec)  
Records: 6  Duplicates: 0  Warnings: 0  
  
mysql> SELECT AVG(Mark) AS Average_Mark  
-> FROM STUDENT7  
-> WHERE CourseNo = 'C002';  
+-----+  
| Average_Mark |  
+-----+  
|      75.0000 |  
+-----+  
1 row in set (0.01 sec)
```

2) List the maximum, minimum mark for “C002”

```
mysql> SELECT  
->     MAX(Mark) AS Maximum_Mark,  
->     MIN(Mark) AS Minimum_Mark  
-> FROM STUDENT7  
-> WHERE CourseNo = 'C002';  
+-----+-----+  
| Maximum_Mark | Minimum_Mark |  
+-----+-----+  
|           90 |           60 |  
+-----+-----+  
1 row in set (0.01 sec)
```

3) List the maximum, minimum, average mark for each course.

```
mysql> SELECT
->     CourseNo,
->     MAX(Mark) AS Maximum_Mark,
->     MIN(Mark) AS Minimum_Mark,
->     AVG(Mark) AS Average_Mark
-> FROM COURSE_MARKS2
-> GROUP BY CourseNo;
```

CourseNo	Maximum_Mark	Minimum_Mark	Average_Mark
C001	90	75	83.3333
C002	80	60	70.0000
C003	95	65	81.6667
C004	88	72	79.3333

4 rows in set (0.01 sec)

4) List the name of the courses and average mark of each course.

```
mysql> SELECT CourseNo,
->     AVG(Mark) AS Average_Mark
-> FROM COURSE_MARKS2
-> GROUP BY CourseNo;
```

CourseNo	Average_Mark
C001	83.3333
C002	70.0000
C003	81.6667
C004	79.3333

4 rows in set (0.02 sec)

5) Calculate the sum of all the scores.

```
mysql> SELECT SUM(Mark) AS Total_Score
-> FROM COURSE_MARKS2;
```

Total_Score
943

1 row in set (0.00 sec)

6) How many students are registered for each course? Display the course description and the number of students registered in each course

```
mysql> SELECT CourseDesc,  
-> COUNT(Reg_No) AS Number_of_Students  
-> FROM COURSE_STUDENTS  
-> GROUP BY CourseNo, CourseDesc;
```

CourseDesc	Number_of_Students
Database	3
Networks	3
Operating Systems	2
Python	4

4 rows in set (0.00 sec)

7) How many courses did each student register for?

```
mysql> SELECT StudentName,  
-> COUNT(CourseNo) AS Number_of_Courses  
-> FROM COURSE_STUDENTS  
-> GROUP BY Reg_No, StudentName;
```

StudentName	Number_of_Courses
Arun	1
Divya	1
Karthik	1
Mohan	1
Priya	1
Ram	1
Anitha	1
Rahul	1
Vijay	1
Kiran	1
Sanjay	1
Meena	1

12 rows in set (0.00 sec)

RESULT:

The records from the tables are displayed using SELECT commands with WHERE Clause and Pattern matching.